CAT Algebra Formula & Practice Sheet

Quadratic & Higher Order Equations, Modulus & Inequalities, Function Transformations & Logarithm Rules

1. Core Algebraic Formulas & Identity Cheatsheet

High-frequency algebraic identities tested regularly in CAT Quant.

Algebraic Domain	Core Mathematical Identity / Theorem	CAT Exam Application Note
Polynomial Expansions	(a+b+c)^2 = a^2+b^2+c^2 + 2(ab+bc+ca)	Used in symmetrical equation solving
Cubic Symmetrical Identity	a^3+b^3+c^3 - 3abc = (a+b+c)(a^2+b^2+c^2-ab-bc-ca)	If a+b+c = 0, then a^3+b^3+c^3 = 3abc (frequent CAT trap)
Roots of Quadratic Equations	Sum: ± + ² = - b / a | Product: ± ² = c / a	Forms basis of 80% of quadratic CAT questions
Modulus Functions	|x - a| represents distance from point a on number line	Split into piecewise domains: x >= a and x < a
Arithmetic-Geometric Mean	AM >= GM for all positive real numbers: (a+b)/2 > = " (a b)	Instant shortcut for finding minimum / maximum values
Logarithmic Identities	log_b(a) = log_c(a)/log_c(b) | a^(log_b c) = c^(log_b a)	Always verify base constraints: b > 0, 0

STRATEGIC PRINCIPLE: Strategic Importance: Algebra typically accounts for 6 to 8 questions in CAT QA alongside Arithmetic.

2. High-Yield Algebra Problem-Solving Strategies

Master these non-standard techniques to solve complex algebraic equations in under 90 seconds.

Strategy Name	When to Apply Technique	Step-by-Step Execution Protocol
Value Substitution	Questions with universal variables (for all x, y > 0)	Substitute small prime integers (x=1, y=2) and evaluate options
Graphical Intersection	Equations involving |x|, quadratics, or ceiling functions	Sketch rough graphs of LHS and RHS; count intersection points
Wavy Curve (Sign Scheme)	Rational inequalities: (x-a)(x-b)/(x-c) >= 0	Plot roots on number line; alternate signs from right to left
Symmetry Exploitation	Cyclic expressions involving a, b, c variables	Assume a = b = c = 1 to test expressions for constant invariance

STRATEGIC PRINCIPLE: Warning: Watch for extraneous roots when squaring both sides of radical equations.

3. 4-Week Algebra Mastery Roadmap

Sequenced progression to build confidence from linear equations to advanced function composites.

- Week 1: Linear Equations, Simultaneous Systems & Special Integral Solutions.
- Week 2: Quadratic Equations, Nature of Roots, Maximum/Minimum & Wavy Curve Inequalities.
- Week 3: Modulus Functions, Absolute Values, Logarithms, Surds & Indices.
- Week 4: Functions & Graphs, Composite Functions f(g(x)), Sequence & Series (AP, GP, HP).